Ali Alfatemi

Ph.D. Candidate in Computer Science · Teaching Fellow

New York, NY
aalfatemi@fordham.edu
alialfatemi.github.io
Google Scholar · ORCID
GitHub · LinkedIn

RESEARCH PROFILE

Computer Science Ph.D. candidate at Fordham University developing robust, data-efficient AI systems for network security and visual intelligence. First-author research spans lightweight, noise-tolerant, few-shot, fusion-based, and LLM-assisted DDoS detection for edge and IoT environments; foreground-centric fine-grained recognition; and vision–language generation. Earlier work applies manifold learning to multi-omics cancer subtyping.

258	10	10	13
CITATIONS	H-INDEX	I10-INDEX	FIRST-AUTHOR WORKS

Google Scholar metrics accessed July 30, 2026. Curated record: 22 published papers and 5 preprints; duplicate versions consolidated.

NETWORK SECURITY

Lightweight, noise-tolerant, few-shot, fusion-based, and LLM-assisted DDoS detection.

VISUAL INTELLIGENCE

Foreground-centric recognition, fine-grained classification, and vision–language generation.

COMPUTATIONAL BIOMEDICINE

Multi-omics cancer subtyping on Grassmann manifolds and deep-learning medical image segmentation.

EDUCATION

2022–May 2027 (expected) — **Ph.D. Candidate in Computer Science · Fordham University**
New York, NY · AI for network security, robust and data-efficient learning, computer vision, and multimodal AI. Available for academic and industry research roles beginning May 2027.

2018–2021 — **M.S. in Computer Science · South China University of Technology**
Guangzhou, China · Thesis: *Cancer subtyping by integration of multiomics data on the Grassmann manifold.*

2008–2012 — **B.S. in Computer Science · Thamar University**
Dhamar, Yemen · Thesis: *An Analyzer for Secure Information Flow in Android Applications.*

ACADEMIC & RESEARCH APPOINTMENTS

2025–Present — **Teaching Fellow · Fordham University**
Teaching and mentoring in Computer and Information Science, including course instruction, laboratories, grading, office hours, and student support.

Summer 2025 — **AI Summer Research Fellow · Fordham University**
Research on LLM-enhanced intrusion detection and robust, data-efficient machine learning for advanced network threats.

2021–2022 — **Student Research Intern · Tsinghua Shenzhen International Graduate School**
Investigated deep-learning methods for mitochondria segmentation in medical images.

TEACHING EXPERIENCE

2025–2026 — **CISC 1100: Structures of Computer Science · Fordham University**
Teaching Fellow for Fall 2025 and Spring 2026 offerings. Topics include sets, logic, Boolean algebra, relations, functions, recursion, combinatorics, graph theory, and computer-based laboratory work.

ADDITIONAL EXPERIENCE

2023 — **Professional Development Intern · IBM**
Professional training in applied AI, cloud technologies, and enterprise innovation.

FIRST-AUTHOR PUBLISHED PAPERS · 12

Reverse chronological order. Publisher and DOI records take precedence over duplicate or preliminary index entries.

1. **Ali Alfatemi**, Mohamed Rahouti, Abdellah Chehri, and Zakirul Alam Bhuiyan. “ShallowNet: A Lightweight Neural Network Approach for Efficient Flow-Level DDoS Detection.” IEEE Transactions on Network and Service Management, vol. 23, pp. 5940–5949, 2026. DOI
2. **Ali Alfatemi**, Mohamed Rahouti, Mohammed Aledhari, Nasir Ghani, Abdellah Chehri, and Gwanggil Jeon. “Vision-Language Integration for Image Captioning Using Vision Transformers and GPT-J.” AIPR 2025, Lecture Notes in Computer Science 16446, pp. 101–114, 2026. DOI
3. **Ali Alfatemi**, Mohamed Rahouti, Zakirul Alam Bhuiyan, Abdellah Chehri, and Aiman Solyman. “A Two-Stage LLM-Enhanced DDoS Detection Framework for Next-Generation IoT and Edge Networks.” IEEE GLOBECOM, CISS Symposium, pp. 7–12, 2025. DOI
4. **Ali Alfatemi**, Mohamed Rahouti, Majjed Al-Qatf, and Senthil Kumar Jagatheesaperumal. “Foreground-Centric learning improves robustness in fine-grained visual recognition.” Signal, Image and Video Processing, vol. 19, issue 14, article 1178, 2025. DOI
5. **Ali Alfatemi**, Mohamed Rahouti, Zakirul Alam Bhuiyan, Abdellah Chehri, Mohammed Aledhari, and Gwanggil Jeon. “Enhancing DDoS Detection for Edge Networks With Noise-Tolerant Shallow Neural Networks in Computational Social Systems.” IEEE Transactions on Computational Social Systems, pp. 1–14, 2025. DOI
6. **Ali Alfatemi**, Mohamed Rahouti, Zakirul Alam Bhuiyan, Aiman Solyman, and Mohammed Aledhari. “ProtoMAML: A Hybrid Meta-Learning Approach Integrating Prototypical Networks for Data-Efficient DDoS Attack Detection.” IWCMC, pp. 1144–1149, 2025. DOI
7. **Ali Alfatemi**, Mohamed Rahouti, D. Frank Hsu, Christina Schweikert, Nasir Ghani, Aiman Solyman, and Mohammad I. Saryuddin Assaqty. “Identifying Distributed Denial of Service Attacks Through Multi-Model Deep Learning Fusion and Combinatorial Analysis.” Journal of Network and Systems Management, vol. 33, issue 1, article 8, 2025. DOI
8. **Ali Alfatemi**, Diogo Oliveira, Mohamed Rahouti, Abdelatif Hafid, and Nasir Ghani. “Precision DDoS Detection Through Gaussian Noise-Augmented Neural Networks.” 15th International Conference on Network of the Future, pp. 178–185, 2024. DOI
9. **Ali Alfatemi**, Mohamed Rahouti, D. Frank Hsu, and Christina Schweikert. “Advancing NCAA March Madness Forecasts Through Deep Learning and Combinatorial Fusion Analysis.” IntelliSys, Lecture Notes in Networks and Systems 1067, pp. 539–560, 2024. DOI
10. **Ali Alfatemi**, Sarah A. L. Jamal, Nasim Paykari, Mohamed Rahouti, and Abdellah Chehri. “Multi-Label Classification with Deep Learning and Manual Data Collection for Identifying Similar Bird Species.” Procedia Computer Science, vol. 246, pp. 558–565, 2024. DOI
11. **Ali Alfatemi**, Sarah A. L. Jamal, Nasim Paykari, Mohamed Rahouti, Ruhul Amin, and Abdellah Chehri. “Refining Bird Species Identification Through GAN-Enhanced Data Augmentation and Deep Learning Models.” Procedia Computer Science, vol. 246, pp. 548–557, 2024. DOI
12. **Ali Alfatemi**, Hong Peng, Wentao Rong, Bin Zhang, and Hongmin Cai. “Patient subgrouping with distinct survival rates via integration of multiomics data on a Grassmann manifold.” BMC Medical Informatics and Decision Making, vol. 22, article 190, 2022. DOI

RESEARCH SOFTWARE

warbler_yolo · End-to-end fine-grained vision pipeline

Open-source workflow that segments birds from unannotated field images, creates standardized foreground crops and reproducible data splits, and trains and evaluates a YOLO11 classifier. Released with the peer-reviewed foreground-centric recognition study.

Python · PyTorch · Ultralytics YOLO11 · OpenCV · scikit-learn · Associated paper

INTELLECTUAL PROPERTY

Granted 2023 — **Inventor · CN113537358B**

A method and system for cancer subtype identification using multi-omics data, with Hongmin Cai. Patent record.

PROFESSIONAL SERVICE

Journal reviewer — Information Fusion; IEEE Transactions on Network and Service Management; Scientific Reports; Signal, Image and Video Processing; Artificial Intelligence Review; The Journal of Supercomputing; Cluster Computing.

Conference reviewer — IEEE World Congress on Computational Intelligence 2024; International Joint Conference on Neural Networks 2025.

ADDITIONAL PUBLISHED PAPERS · 10

1. Mohammad Iqbal Saryuddin Assaqty, Ying Gao, Xiping Hu, **Ali Alfatemi**, Handy Fernandy, Ircham Ali, and Peihao Zhang. "XManuChain: A Permissioned Blockchain Framework for Securing IoP Transactions in Production Process Collaboration." Journal of Network and Systems Management, vol. 34, issue 4, article 119, 2026. DOI
2. Aiman Solyman, Ahmed Elazab, Mohamed Rahouti, **Ali Alfatemi**, Zeinab Mahmoud, and Marco Zappatore. "MS-GBANet: Multiscale graph convolution with boundary attention for medical image segmentation." Biomedical Signal Processing and Control, vol. 121, article 110397, 2026. DOI
3. Zeinab Mahmoud, Marco Zappatore, Natalia Kryvinska, Mohammed Abdalsalam, **Ali Alfatemi**, and Aiman Solyman. "MTAGEC: Multi-Task Arabic Grammatical Error Correction as a sequence generation with synthetic data." Journal of King Saud University Computer and Information Sciences, vol. 38, issue 2, article 35, 2026. DOI
4. Mohammad Iqbal Saryuddin Assaqty, Ying Gao, **Ali Alfatemi**, Mohamed Rahouti, and Abdellah Chehri. "A Resource-Efficient Blockchain With Delegated Fault-Tolerance for Manufacturing Nodes." IEEE Transactions on Industrial Informatics, vol. 22, issue 5, pp. 4340–4349, 2026. DOI
5. Henok Wondimu, **Ali Alfatemi**, Mohamed Rahouti, Abdellah Chehri, Paweł Weichbroth, and Nasir Ghani. "Enabling Real-Time, Explainable DDoS Mitigation via On-Premise Large Language Models and Flow Analysis." Procedia Computer Science, vol. 270, pp. 1390–1399, 2025. DOI
6. Zeinab Mahmoud, Natalia Kryvinska, Mohammed Abdalsalm, Aiman Solyman, **Ali Alfatemi**, and Ahmad Musyafa. "Toward Utilizing Bidirectional Multi-head Attention Technique for Automatic Correction of Grammatical Errors." IAENG International Journal of Computer Science, vol. 51, issue 11, pp. 1886–1897, 2024.
7. Senthil Kumar Jagatheesaperumal, Mohamed Rahouti, **Ali Alfatemi**, Nasir Ghani, Vu Khanh Quy, and Abdellah Chehri. "Enabling Trustworthy Federated Learning in Industrial IoT: Bridging the Gap Between Interpretability and Robustness." IEEE Internet of Things Magazine, vol. 7, issue 5, pp. 38–44, 2024. DOI
8. Fernando Martinez, Mariyam Mapkar, **Ali Alfatemi**, Mohamed Rahouti, Yufeng Xin, Kaiqi Xiong, and Nasir Ghani. "Redefining DDoS Attack Detection Using a Dual-Space Prototypical Network-Based Approach." ICCCN, pp. 1–9, 2024. DOI
9. Zeinab Mahmoud, Chunlin Li, Marco Zappatore, Aiman Solyman, **Ali Alfatemi**, Ashraf Osman Ibrahim, and Abdelzahir Abdelmaboud. "Semi-Supervised Learning and Bidirectional Decoding for Effective Grammar Correction in Low-Resource Scenarios." PeerJ Computer Science, vol. 9, e1639, 2023. DOI
10. Aiman Solyman, Marco Zappatore, Wang Zhenyu, Zeinab Mahmoud, **Ali Alfatemi**, Ashraf Osman Ibrahim, and Lubna Abdelkareim Gabralla. "Optimizing the Impact of Data Augmentation for Low-Resource Grammatical Error Correction." Journal of King Saud University – Computer and Information Sciences, vol. 35, issue 6, article 101572, 2023. DOI

PREPRINTS · 5

1. PREPRINT Nasim Paykari, **Ali Alfatemi**, Damian M. Lyons, and Mohamed Rahouti. "Integrating Robotic Navigation with Blockchain: A Novel PoS-Based Approach for Heterogeneous Robotic Teams." Conference presentation, IEEE UR 2024; arXiv:2505.15954, 2025. arXiv
2. PREPRINT Mohammed Aledhari, Mohamed Rahouti, and **Ali Alfatemi**. "Towards Equitable ASD Diagnostics: A Comparative Study of Machine and Deep Learning Models Using Behavioral and Facial Data." arXiv:2411.05880, 2024. arXiv
3. PREPRINT Nasim Paykari, Razieh Jokar, **Ali Alfatemi**, Damian Lyons, and Mohamed Rahouti. "Optimizing Control Strategies for Wheeled Mobile Robots Using Fuzzy Type I and II Controllers and Parallel Distributed Compensation." arXiv:2409.17161, 2024. arXiv
4. PREPRINT **Ali Alfatemi**, Mohamed Rahouti, Ruhul Amin, Sarah ALJamal, Kaiqi Xiong, and Yufeng Xin. "Advancing DDoS Attack Detection: A Synergistic Approach Using Deep Residual Neural Networks and Synthetic Oversampling." arXiv:2401.03116, 2024. arXiv
5. PREPRINT Mohammad Iqbal Saryuddin Assaqty, Ying Gao, **Ali Alfatemi**, Ashraf Osman Ibrahim, Anas W. Abulfaraj, Faisal Binzagr, Fezan Nabawi, and Mohammad Reza Fahlevi. "PBEMS: A Permissioned Blockchain-based Equipment Maintenance System in Smart Manufacturing." Preprints.org, version 2, 2023. DOI

Complete, maintained bibliography: alialfatemi.github.io/publications/

TECHNICAL EXPERTISE

Programming	Python · C/C++ · Bash
Machine learning	PyTorch · TensorFlow · scikit-learn
Computer vision	OpenCV · YOLO/Ultralytics · TorchVision
Research systems	Git · Linux · HPC and batch workflows · large-scale data handling
Methods	Data augmentation · GANs · robust learning · meta-learning · model evaluation